Steam in a test tube

Y17 (2017) — English translation

A thin test tube is partially filled with water and placed vertically in a large volume chamber in which the air is maintained at zero relative humidity and a constant pressure equal to external atmospheric pressure. Due to diffusion in the test tube, a linear change in the vapor concentration with height is established: near the surface of the water, the vapor turns out to be saturated, and at the upper open end of the test tube its concentration is 2 times less. The top of the test tube is closed with a lid. After equilibrium was established, it turned out that the density of moist air inside the test tube differs from the density of dry air away from the test tube by $\Delta\rho = 5.4$ g/m^3. Molar mass of dry air $\mu_{\text{in}} = 29$ g/mol, vapor $\mu_{\text{p}} = 18$ g/mol. Neglect the change in the liquid level in the test tube during the experiment.

A1	Find the saturated vapor pressure at the experimental temperature equal to $T = 27°\text{C}$.	**3.00**

Source: `https://pho.rs/p/3036`